

Highlights

Spectral Co-Design of Agrivoltaic Modules for Multifunctional Net Energy Benefit: A Quantum-Guided Digital Twin

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- Dual-band filtering at 750/820 nm extends FMO coherence by 50% at 295 K.
- NPoM cavity enables SERS crop diagnostics; transport trade-off quantified.
- OPV shield cuts evapotranspiration 28% (FAO-56); NEB 72.6 kgCO₂e/m²/yr.
- 500 m² cooperative module pays back in 4.32 yr in Cameroon highlands.
- First, to our knowledge, coherence-guided agrivoltaic design with TEA and LCA.

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Abstract

Spectral co-design offers a route beyond the system-level trade-off of static agrivoltaics, in which semi-transparent modules resolve the food–energy land conflict only by uniformly sacrificing electrical output or crop productivity. Here we present a quantum-guided agrivoltaic stack in which a semi-transparent organic photovoltaic layer couples to a plasmonic nanoparticle-on-mirror (NPoM) cavity that actively shapes the transmitted spectrum. By transmitting narrow bands at 750 nm and 820 nm, the system prepares polaron-dressed states in the Fenna–Matthews–Olson photosynthetic complex and extends exciton coherence by 50 %; confining the plasmonic sentinel layer to 1 % of the area then preserves 97.1 % of the device-level excitation-transfer yield (a device metric, not crop yield) at a 91.8 % local transport penalty on the sentinel. The NPoM sentinel layer enables multiplexed SERS and quantum dot diagnostics of agricultural contaminants and crop stress. On the macroscopic scale, the shield reduces crop evapotranspiration by 28 % and yields a Net Ecological Benefit of $72.6 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$. A BB84 quantum link secures the IoT telemetry. Finally, techno-economic analysis of a Cameroonian micro-module gives a payback period of 4.32 yr, identifying conditions under which the multifunctional net benefit stays positive for the modeled 500 m^2 cooperative case.

Keywords: agrivoltaic spectral management, net ecological benefit, FAO-56 microclimate, plasmonic nanoparticle-on-mirror cavity, SERS diagnostics,

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techno-economic analysis, quantum-guided digital twin
